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Exercises

1. Find the first three nonzero terms of the Maclaurin series for the following functions:

- (a) $y = e^{-2x}$ (b) $y = \log(\sec x)$
(c) $y = \log(\cos x)$ (d) $y = \sec x$
(e) $y = \cos^2 x$ (f) $y = \tan^{-1}(1+x)$
(g) $y = \log(1+e^x)$ (h) $y = \log(1+\sin x)$
(i) $y = e^x \cos x$

2. Expand $f(x) = \frac{1}{x}$ in powers of $(x-2)$.

3. Expand $e^{\sin x} \sin(px)$ by Maclaurin's theorem with Cauchy's form of remainders.

4. Expand $\log(1+x)$ by Maclaurin's theorem with Lagrange's form of remainders after $2n$ terms.

5. Prove that (i) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n+1}}{(2n+1)!} \sin(\theta x)$, $0 < \theta < 1$

(ii) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots - \frac{x^{n-1}}{n-1} - \frac{x^n}{n(1-\theta x)^n}$, $0 < \theta < 1$

6. Expand $\log(1+x)$ by Taylor's theorem with Lagrange's form of remainders.

Maclaurin's Theorem $\Rightarrow f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$

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Soln: (a)

$f(x) = y = e^{-2x} \Rightarrow f(0) = e^{-2 \cdot 0} = 1$

$f'(x) = y_1 = -2e^{-2x} \Rightarrow f'(0) = -2e^{-2 \cdot 0} = -2$

$f''(x) = y_2 = 4e^{-2x} \Rightarrow f''(0) = 4e^{-2 \cdot 0} = 4$

$f'''(x) = y_3 = -8e^{-2x} \Rightarrow f'''(0) = -8e^{-2 \cdot 0} = -8$

$\therefore f(x) = e^{-2x} = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

$= 1 + \frac{x}{1!}(-2) + \frac{x^2}{2!} \cdot 4 + \frac{x^3}{3!}(-8) + \dots$

$= 1 - 2x + 2x^2 - \frac{4}{3}x^3 + \dots$

Soln:

(b) $f(x) = y = \log(\sec x) \Rightarrow f(0) = 0$

$f'(x) = y_1 = \frac{\sec x \tan x}{\sec x} = \tan x \Rightarrow f'(0) = 0$

$f''(x) = y_2 = \sec^2 x \Rightarrow f''(0) = 1$

$f'''(x) = y_3 = 2\sec^2 x \tan x \Rightarrow f'''(0) = 0$

$f^{(4)}(x) = y_4 = 2\sec^4 x + 4\sec^2 x \tan^2 x \Rightarrow f^{(4)}(0) = 2$

$f^{(5)}(x) = y_5 = 8\sec^4 x \tan x + 8\sec^2 x \tan^3 x \Rightarrow f^{(5)}(0) = 0$

$f^{(6)}(x) = y_6 = 8\sec^6 x + 22\sec^4 x \tan^2 x + 16\sec^2 x \tan^4 x \Rightarrow f^{(6)}(0) = 8 + 8 = 16$

$\therefore f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

$= 0 + \frac{x}{1!} \cdot 0 + \frac{x^2}{2!} \cdot 1 + \frac{x^3}{3!} \cdot 0 + \frac{x^4}{4!} \cdot 2 + \frac{x^5}{5!} \cdot 0 + \frac{x^6}{6!} \cdot 16 + \dots$

$= \frac{x^2}{2} + \frac{x^4}{12} + \frac{x^6}{45} + \dots$

(c.d.e $\Rightarrow$ same type $\rightarrow$ keep derivating until you find first three non zero terms)

Soln:

(f) $f(x) = y = \tan^{-1}(1+x)$

$\Rightarrow f(0) = \frac{\pi}{4}$

$$f'(x) = y_1 = \frac{1}{\{1+(1+x)^2\}} \Rightarrow f'(0) = \frac{1}{2}$$

$$f''(x) = y_2 = -\frac{1}{\{1+(1+x)^2\}^2} \cdot 2(1+x) \cdot 1 = \frac{-2(1+x)}{\{1+(1+x)^2\}^2} \Rightarrow f''(0) = \frac{-2}{2^2} = -\frac{1}{2}$$

$$f'''(x) = y_3 = \frac{2}{\{1+(1+x)^2\}^3} \cdot 2(1+x) \cdot 1 + \frac{4(1+x)}{\{1+(1+x)^2\}^3} \Rightarrow f'''(0) = \frac{-4}{2^3} = -\frac{1}{2}$$

$$\therefore f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$= \frac{x}{1} + \frac{x}{1!} \cdot \frac{1}{2} + \frac{x^2}{2!} \left(-\frac{1}{2}\right) + \frac{x^3}{3!} \left(-\frac{1}{2}\right) + \dots$$

$$= \frac{x}{1} + \frac{x}{2} - \frac{x^2}{4} + \frac{x^3}{12} + \dots$$

(g, h) $\Rightarrow$ same type $\Rightarrow$ practice on your own $\odot$

- After all you have to answer all the questions $\odot$



Solⁿ: ①

$$f(x) = y = e^x \cos x \Rightarrow f(0) = 1$$

$$f'(x) = y_1 = e^x \cos x - e^x \sin x \Rightarrow f'(0) = 1$$

$$f''(x) = y_2 = e^x \cos x - e^x \sin x - e^x \cos x - e^x \sin x = -2e^x \sin x \Rightarrow f''(0) = 0$$

$$f'''(x) = y_3 = -2e^x \sin x - 2e^x \cos x \Rightarrow f'''(0) = -2$$

$$f^{(4)}(x) = y_4 = -2e^x \sin x - 2e^x \cos x + 2e^x \sin x - 2e^x \cos x = -4e^x \cos x \Rightarrow f^{(4)}(0) = -4$$

$$\therefore f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$= 1 + \frac{x}{1!} \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} \cdot (-2) + \frac{x^4}{4!} \cdot (-4) + \dots$$

$$= 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

Taylor's Theorem $\Rightarrow$

in power of $(x-a)$,

$$f(x) = f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots$$

② $f(x) = \frac{1}{x}$ (in the power of $(x-2)$)

$\therefore f(a) = \frac{1}{2}$ $\because a=2$

$\therefore f'(x) = -\frac{1}{x^2} \Rightarrow f'(a) = -\frac{1}{4}$

$\therefore f''(x) = \frac{2}{x^3} \Rightarrow f''(a) = \frac{2}{2^3} = \frac{1}{4}$

$\therefore f'''(x) = -\frac{6}{x^4} \Rightarrow f'''(a) = -\frac{6}{2^4} = -\frac{3}{8}$

Now, $f(x) = f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f'''(a)}{3!} (x-a)^3 + \dots$

$$= \frac{1}{2} - \frac{1}{4} (x-2) + \frac{1}{8} (x-2)^2 - \frac{1}{16} (x-2)^3 + \dots$$

Taylor's theorem with Cauchy's Remainder $\rightarrow$

If f is a function defined in $[a, b]$ such that

① $f, f', f'', \dots, f^{(n-1)}(x)$ are continuous in $[a, b]$

② $f^{(n)}(x)$ exists in (a, b)

then there exists at least one real number $0 < \theta < 1$, such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(b-a)^n}{(n)!}f^{(n)}(\theta) [a, b]$$

Put, $b = a+h \Rightarrow b-a = h$

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{h^n}{(n)!}f^{(n)}(a+\theta h)$$

(where $0 < \theta < 1$)

Maclaurin's theorem with Cauchy's Remainder: $\Rightarrow$

Put $a=0, h=x$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + \frac{x^n}{(n)!}f^{(n)}(\theta x) \quad 0 < \theta < 1$$

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Here, $f(x) = e^{ax} \sin px \Rightarrow f(0) = 0$

$$\Rightarrow f'(x) = e^{ax}(p \cos px + a \sin px) \Rightarrow f'(0) = p$$

$$\Rightarrow f''(x) = e^{ax}(a^2 \sin px + 2ap \cos px) \Rightarrow f''(0) = 2ap$$

$$\Rightarrow f'''(x) = e^{ax}(a^3 \cos px + (3a^2p - p^3) \sin px) \Rightarrow f'''(0) = 3a^2p - p^3 = p(3a^2 - p^2)$$

$$\vdots$$

$$\Rightarrow f^{(n)}(x) = e^{ax}(a^2 + p^2)^{n/2} x \sin \left[px + n \tan^{-1} \frac{p}{a} \right]$$

$$\Rightarrow f^{(n)}(\theta x) = e^{a\theta x}(a^2 + p^2)^{n/2} x \sin \left(p\theta x + n \tan^{-1} \frac{p}{a} \right)$$

Now,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + \frac{x^n}{(n)!}f^{(n)}(\theta x)$$

$$= 0 + xp + \frac{x^2}{2}p(3a^2 - p^2) + \dots + \frac{x^n}{(n-1)!}(1-\theta)^{n-1} e^{a\theta x}(a^2 + p^2)^{n/2} x \sin \left(p\theta x + n \tan^{-1} \frac{p}{a} \right)$$

$$e^{ax}(p \cos px + a \sin px)$$

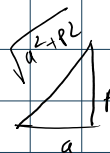
$$\Rightarrow e^{ax} \sqrt{a^2 + p^2} \left(\frac{p}{\sqrt{a^2 + p^2}} \cos px + \frac{a}{\sqrt{a^2 + p^2}} \sin px \right)$$

$$\Rightarrow e^{ax} \sqrt{a^2 + p^2} \left(\sin px \cos \tan^{-1} \frac{p}{a} + \cos px \sin \tan^{-1} \frac{p}{a} \right)$$

$$\Rightarrow e^{ax} \sqrt{a^2 + p^2} \left\{ \sin \left(px + \tan^{-1} \frac{p}{a} \right) \right\}$$

$$\vdots$$

$$\Rightarrow e^{a\theta x}(a^2 + p^2)^{n/2} \sin \left(p\theta x + n \tan^{-1} \frac{p}{a} \right)$$



Taylor's theorem with Lagrange's form of remainder: $\Rightarrow$

If f is a function defined in $[a, b]$ such that

① $f, f', f'', \dots, f^{(n-1)}(x)$ are continuous in $[a, b]$

② $f^{(n)}(x)$ exists in (a, b)

then, there at least exist one real number $c \in (a, b)$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(b-a)^n}{n!}f^{(n)}(c).$$

Maclaurin's theorem with Lagrange's form of remainder: $\Rightarrow$

If f is a function defined in $[a, b]$ such that

① $f, f', f'', \dots, f^{(n-1)}(x)$ are continuous in $[a, b]$

② $f^{(n)}(x)$ exists in (a, b)

(i) $f, f', f'', \dots, f^{(n-1)}(x)$ are continuous in $[a, b]$

(ii) $f^{(n)}(x)$ exists in (a, b)

then, there at least exist one real number $c \in (a, b)$ such that

$$f(b) = f(a) + (b-a)f'(a) + \frac{(b-a)^2}{2!}f''(a) + \dots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(b-a)^n}{n!}f^{(n)}(c).$$

Now, put $b = a+h \Rightarrow (a, a+h)$

$$\Rightarrow (a+\theta h, a+\theta h)$$

We can write $c = a + \theta h$ ($0 < \theta < 1$)

$$\text{So, } f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{h^n}{n!}f^{(n)}(a+\theta h)$$

Now, $a=0, h=x$.

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + \frac{x^n}{n!}f^{(n)}(\theta x)$$

$(0 < \theta < 1)$

4) $\log(1+x)$

$$\Rightarrow f(x) = \log(1+x) \Rightarrow f(0) = 0$$

$$\Rightarrow f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$$

$$\Rightarrow f''(x) = -\frac{1}{(1+x)^2} \Rightarrow f''(0) = -2$$

$$\Rightarrow f'''(x) = \frac{(-1)^2 \cdot 2!}{(1+x)^3} \Rightarrow f'''(0) = 2$$

$$\begin{aligned} f(x) &= f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots + \frac{x^n}{n!}f^{(n)}(0) + \dots + \frac{x^{2n}}{2n!}f^{(2n)}(\theta x) \\ &= 0 + x + \frac{x^2}{2!}(-2) + \frac{x^3}{3!}2 + \dots + \frac{x^n}{n!} \times \frac{(-1)^{n-1}(n-1)!}{(1+\theta x)^n} + \dots + \frac{x^{2n}}{2n!} \times \frac{(-1)^{2n-1}(2n-1)!}{(1+\theta x)^{2n}} \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n(n-1)!} \times \frac{(-1)^{n-1}(n-1)!}{(1+\theta x)^n} \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{(-1)^{n-1}x^n}{n(1+\theta x)^n} + \dots + \frac{(-1)^{2n-1}x^{2n}}{2n(1+\theta x)^{2n}} \end{aligned}$$

5) (i) $f(x) = \sin x \Rightarrow f(0) = 0$

$$\Rightarrow f'(x) = \cos x \Rightarrow f'(0) = 1$$

$$\Rightarrow f''(x) = -\sin x \Rightarrow f''(0) = 0$$

$$\Rightarrow f'''(x) = -\cos x \Rightarrow f'''(0) = -1$$

Now,

$$f(x) = \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n}}{(2n)!} \sin(\theta x), \quad 0 < \theta < 1.$$

Now,

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n+1}}{(2n)!} \sin(\theta x), \quad 0 < \theta < 1.$$

$$= 0 + x + 0 - \frac{x^3}{3!} - 0 + \frac{x^5}{5!} + \dots - \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n+1}}{(2n)!} \sin(\theta x), \quad 0 < \theta < 1.$$

[i] $\Rightarrow$ same as [i]

② $\log(1+x)$

$$\Rightarrow f(x) = \log(1+x) \Rightarrow f(0) = 0$$

$$\Rightarrow f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$$

$$\Rightarrow f''(x) = -\frac{1}{(1+x)^2} \Rightarrow f''(0) = -2$$

$$\Rightarrow f'''(x) = \frac{(-1)^2 2!}{(1+x)^3} \Rightarrow f'''(0) = 2$$

The Taylor series:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

Assuming, $a=0$,

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(\theta x) \quad [\text{With Lagrange's Remainder}]$$

$$= 0 + x + \frac{x^2}{2!} (-2) + \frac{x^3}{3!} 2 + \dots + \frac{x^n}{n!} \times \frac{(-1)^{n-1} (n-1)!}{(1+\theta x)^n}$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n(n-1)!} \times \frac{(-1)^{n-1} (n-1)!}{(1+\theta x)^n}$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{(-1)^{n-1} x^n}{n(1+\theta x)^n}$$

(I am tired 😊)